

```

explain analyze
with category_stats as (
  select
    p.product_category,
    count(*) as orders_count
  from opt_orders o
  join opt_clients c
    on c.id = o.client_id
  join opt_products p
    on p.product_id = o.product_id
  where c.status = 'active'
    and o.order_date >= date '2023-01-01'
    and o.order_date < date '2024-01-01'
  group by p.product_category
)
select
  (
    select product_category
    from category_stats
    order by orders_count desc, product_category asc
    limit 1
  ) as most_popular_category,
  (
    select product_category
    from category_stats
    order by orders_count asc, product_category asc
    limit 1
  ) as least_popular_category,
  (
    select avg(orders_count)
    from category_stats
  ) as avg_orders_per_category

```

QUERY PLAN

Result (cost=10248.74..10248.75 rows=1 width=268) (actual time=480.428..498.876 rows=1.00 loops=1)

Buffers: shared hit=1375

CTE category_stats

-> Finalize GroupAggregate (cost=10246.85..10248.36 rows=5 width=18) (actual time=480.283..498.742 rows=5.00 loops=1)
Group Key: p.product_category
Buffers: shared hit=1375
-> Gather Merge (cost=10246.85..10248.25 rows=12 width=18) (actual time=480.265..498.719 rows=15.00 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=1375
-> Sort (cost=9246.83..9246.84 rows=5 width=18) (actual time=353.501..353.508 rows=5.00 loops=3)
Sort Key: p.product_category
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=1375
Worker 0: Sort Method: quicksort Memory: 25kB
Worker 1: Sort Method: quicksort Memory: 25kB
-> Partial HashAggregate (cost=9246.72..9246.77 rows=5 width=18) (actual time=353.356..353.364 rows=5.00 loops=3)
Group Key: p.product_category
Batches: 1 Memory Usage: 32kB
Buffers: shared hit=1359
Worker 0: Batches: 1 Memory Usage: 32kB
Worker 1: Batches: 1 Memory Usage: 32kB
-> Hash Join (cost=2005.37..9041.10 rows=41124 width=10) (actual time=21.070..333.373 rows=33289.00 loops=3)
Hash Cond: (o.product_id = p.product_id)
Buffers: shared hit=1359

-> Parallel Hash Join (cost=1956.87..8884.19 rows=41124 width=4) (actual time=17.427..310.491 rows=33289.00 loops=3)
Hash Cond: (o.client_id = c.id)
Buffers: shared hit=1241
-> Parallel Index Only Scan using idx_opt_orders_date_client_product on opt_orders o (cost=0.42..6711.90 rows=82220 width=20) (actual time=0.936..233.667 rows=66568.33 loops=3)
Index Cond: ((order_date >= '2023-01-01'::date) AND (order_date < '2024-01-01'::date))
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=991
-> Parallel Hash (cost=1695.95..1695.95 rows=20840 width=16) (actual time=16.034..16.036 rows=16674.67 loops=3)
Buckets: 65536 Batches: 1 Memory Usage: 2880kB
Buffers: shared hit=250
-> Parallel Index Only Scan using idx_opt_clients_status_id on opt_clients c (cost=0.42..1695.95 rows=20840 width=16) (actual time=0.099..16.590 rows=50024.00 loops=1)
Index Cond: (status = 'active'::text)
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=250
-> Hash (cost=36.00..36.00 rows=1000 width=14) (actual time=3.594..3.594 rows=1000.00 loops=3)
Buckets: 1024 Batches: 1 Memory Usage: 55kB
Buffers: shared hit=118
-> Seq Scan on opt_products p (cost=0.00..36.00 rows=1000 width=14) (actual time=2.628..3.169 rows=1000.00 loops=3)

Buffers: shared hit=118
InitPlan 2
-> Limit (cost=0.13..0.13 rows=1 width=126) (actual time=480.343..480.345 rows=1.00 loops=1)
Buffers: shared hit=790
-> Sort (cost=0.13..0.14 rows=5 width=126) (actual time=480.340..480.341 rows=1.00 loops=1)
Sort Key: category_stats.orders_count DESC, category_stats.product_category
Sort Method: top-N heapsort Memory: 25kB
Buffers: shared hit=790
-> CTE Scan on category_stats (cost=0.00..0.10 rows=5 width=126) (actual time=480.289..480.314 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=790
InitPlan 3
-> Limit (cost=0.13..0.13 rows=1 width=126) (actual time=0.056..0.058 rows=1.00 loops=1)
-> Sort (cost=0.13..0.14 rows=5 width=126) (actual time=0.054..0.055 rows=1.00 loops=1)
Sort Key: category_stats_1.orders_count, category_stats_1.product_category
Sort Method: top-N heapsort Memory: 25kB
-> CTE Scan on category_stats category_stats_1 (cost=0.00..0.10 rows=5 width=126) (actual time=0.001..0.003 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
InitPlan 4
-> Aggregate (cost=0.12..0.12 rows=1 width=32) (actual time=0.019..0.020 rows=1.00 loops=1)
-> CTE Scan on category_stats category_stats_2 (cost=0.00..0.10 rows=5 width=8) (actual time=0.001..0.002 rows=5.00 loops=1)

Storage: Memory Maximum Storage: 17kB
Planning:
Buffers: shared hit=11
Planning Time: 2.947 ms
Execution Time: 499.451 ms